

EX NO : 10

DATE : 05-08-26

CROSS VALIDATION

AIM:

To build a classification model using Logistic Regression on the Titanic dataset and evaluate its performance and generalization capability using **K-Fold Cross-Validation**.

PROBLEM STATEMENT:

The **Titanic Dataset** contains passenger details used to predict survival (0 = Died, 1 = Survived) using **8 features**: pclass (ticket class), sex, age, sibsp (siblings/spouses), parch (parents/children), fare, and embarked (boarding port).

CODE:

```
# Import libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import KFold
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load Dataset
df = sns.load_dataset('titanic')

print("First 5 rows")
print(df.head())

print("\nDataset Shape:", df.shape)

# Select useful features
df = df[['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare', 'embarked']]

# Handle Missing Values
df['age'] = df['age'].fillna(df['age'].median())
df['embarked'] = df['embarked'].fillna(df['embarked'].mode()[0])

# Convert Categorical Columns
encoder = LabelEncoder()
df['sex'] = encoder.fit_transform(df['sex'])
df['embarked'] = encoder.fit_transform(df['embarked'])

# Split Features and Target
X = df.drop('survived', axis=1)
y = df['survived']

# Define K-Fold Cross Validation
k = 5
kf = KFold(n_splits=k, shuffle=True, random_state=42)

# Lists to store results per fold
fold_accuracies = []
y_true_all = []
```

```

y_pred_all = []

# Train Logistic Regression Model using K-Fold Loop
model = LogisticRegression(max_iter=1000)

print("\n--- Fold-wise Accuracy Results ---")
for fold, (train_idx, test_idx) in enumerate(kf.split(X, y), start=1):
    # Split data for current fold
    X_train, X_test = X.iloc[train_idx], X.iloc[test_idx]
    y_train, y_test = y.iloc[train_idx], y.iloc[test_idx]

    # Train Model
    model.fit(X_train, y_train)

    # Predictions
    y_pred = model.predict(X_test)

    # Accuracy for the current fold
    acc = accuracy_score(y_test, y_pred) * 100
    fold_accuracies.append(acc)

    # Store labels for overall matrix & report
    y_true_all.extend(y_test)
    y_pred_all.extend(y_pred)

    print(f"Fold {fold} Accuracy : {round(acc, 2)} %")

# Overall Accuracy
mean_acc = np.mean(fold_accuracies)
std_acc = np.std(fold_accuracies)

print("\nAverage K-Fold Accuracy :", round(mean_acc, 2), "%")
print("Standard Deviation      :", round(std_acc, 2), "%")

# Confusion Matrix
print("\nConfusion Matrix")
print(confusion_matrix(y_true_all, y_pred_all))

# Classification Report
print("\nClassification Report")
print(classification_report(y_true_all, y_pred_all))

# Accuracy Graph across Folds
folds = [f"Fold {i}" for i in range(1, k + 1)]

plt.figure(figsize=(7, 5))
plt.bar(folds, fold_accuracies, color='skyblue', edgecolor='black')

# Draw line for mean accuracy
plt.axhline(y=mean_acc, color='red', linestyle='--', label=f'Mean Acc: {mean_acc:.2f}%')

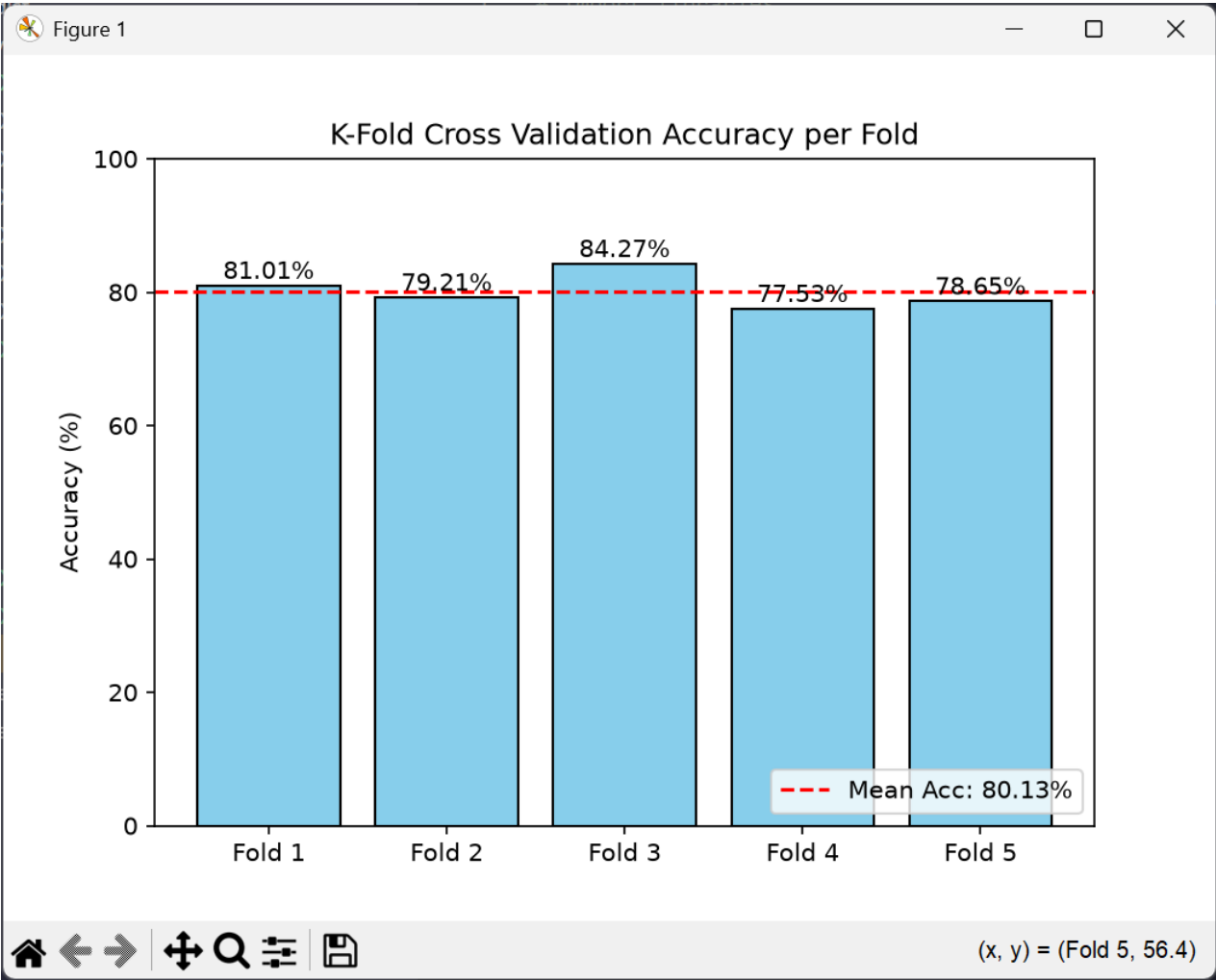
plt.ylabel("Accuracy (%)")
plt.title("K-Fold Cross Validation Accuracy per Fold")
plt.ylim(0, 100)
plt.legend(loc='lower right')

for i, v in enumerate(fold_accuracies):
    plt.text(i, v + 1, f'{v:.2f}%', ha='center', fontsize=10)

plt.show()

```

OUTPUT:



```
First 5 rows
survived  pclass    sex  age  sibsp  parch    fare  ...  class  who  adult_male  deck  embark_town  alive
0         0        3  male  22.0    1     0   7.2500  ...  Third  man      True    NaN  Southampton   no
1         1        1 female  38.0    1     0  71.2833  ...  First  woman   False     C   Cherbourg   yes
2         1        3 female  26.0    0     0   7.9250  ...  Third  woman   False    NaN  Southampton   yes
3         1        1 female  35.0    1     0  53.1000  ...  First  woman   False     C   Southampton   yes
4         0        3  male  35.0    0     0   8.0500  ...  Third  man     True    NaN  Southampton   no

[5 rows x 15 columns]

Dataset Shape: (891, 15)

--- Fold-wise Accuracy Results ---
Fold 1 Accuracy : 81.01 %
Fold 2 Accuracy : 79.21 %
Fold 3 Accuracy : 84.27 %
Fold 4 Accuracy : 77.53 %
Fold 5 Accuracy : 78.65 %

Average K-Fold Accuracy : 80.13 %
Standard Deviation      : 2.35 %
```

Confusion Matrix

```
[[472  77]
 [100 242]]
```

Classification Report

	precision	recall	f1-score	support
0	0.83	0.86	0.84	549
1	0.76	0.71	0.73	342
accuracy			0.80	891
macro avg	0.79	0.78	0.79	891
weighted avg	0.80	0.80	0.80	891